Statistics
Parameter Estimation, Hypothesis Testing, Regression

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CMInDS, EE
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Lecture 2

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Gentle Reminder About Expectations

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- Maintain a notebook and take notes. It helps a lot.
- Try to revise the material at the first opportunity after the lecture. Make the opportunity
- Learning takes time. Last minute cramming is neither efficient nor productive.
- Learning is a hard on the brain and possibly also on the body You have to bear the pain and make the effort.
- Exam efficiency is incidental; not the goal of the course.
- The course is **demanding, but not cruel.**

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Preview of the Rest of the Course

- Three key topics: Estimation, Hypothesis testing, and Regression.
- Estimation: Much like what we saw in the case studies; provide more detailed descriptions:

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Preview of the Rest of the Course

- Three key topics: Estimation, Hypothesis testing, and Regression.
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Preview of the Rest of the Course

- Three key topics: Estimation, Hypothesis testing, and Regression.
- Estimation: Much like what we saw in the case studies; provide more detailed descriptions: *best* point estimates using different criteria, confidence intervals and interval estimates.
- Hypothesis testing: Make claim, formulate an statistical experiment, accept or reject the claim, and specify how sure you are about this.
- Regression: propose a relation between a set of explanatory variables and 'response variable'

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Preview of the Rest of the Course

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- Estimation: Much like what we saw in the case studies; provide more detailed descriptions: *best* point estimates using different criteria, confidence intervals and interval estimates.
- Hypothesis testing: Make claim, formulate an statistical experiment, accept or reject the claim, and specify how sure you are about this.
- Regression: propose a relation between a set of explanatory variables and 'response variable' determine parameters of the model and/or test hypotheses about the relation.
- Key concepts to be used: pdf and pmf, Independent random variables, moments, CLT and LLN to provide some intuition.

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Recap of Probability Concepts

- Properties of probability measure
- Inclusion-exclusion, union bound and Bonferroni inequalities,
- Conditional probability and its properties: chain rule, law of total probability, and Bayes' rule,
- Independence of events
- Counting and sampling: with(out) replacement and with(out) ordering
- Random Variables and CDF
- Discrete Random Variables and PMF
- Continuous Random Variables and PDF

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Recap of Probability Concepts

- Expectation and Variance of random variables, and their properties
- Random vectors and joint CDF, independence of random variables
- Discrete Random Vectors and joint / conditional PMF
- Jointly Continuous Random Vectors and joint / conditional PDF
- Covariance and correlation coefficient
- Much of the material is covered in the text *Introduction to Probability and Statistics for Engineers and Scientists,* by Sheldon Ross.

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One-Slide Revision

- Probability theory is an abstract stand-alone theory developed from the three axioms
- Provides a consistent framework for quantifying "beliefs" about events of interests.
- Inspired by 'real life' but the theory itself is 'stand alone'
- One source of developing intuition that is provided by the theory is the Law of Large Numbers
- It connects the theory to help provide a frequentist's view.

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Gaussian Distribution

- Recall that the probability density function of a Gaussian random variable Z with mean μ and variance σ^2 is

$$f_Z(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$

- $\frac{Z-\mu}{\sigma}$ has zero mean and unit variance.
- Multivariate Gaussians have a nice structure.
- Linear transformations of Gaussian random variables results in a Gaussian random variables.
- Uncorrelated Gaussian random variables are independent
- Conditional expectation of a Gaussian is linear

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Law of Large Numbers

- Let X_1 be a random variable with mean (expectation, average) μ and variance σ^2.
- Let X_i for $i = 1, \ldots, n$ be n samples of the random variable.
- Sample sum and sample mean from n samples is

$$\begin{aligned} S_n &= \sum_{i=1}^{n} \mathsf{X}_n \\ M_n &= \frac{S_n}{n} \end{aligned}$$

- Important: Different realisations of n samples will result in different M_n.

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Law of Large Numbers

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$$\begin{aligned} S_n &= \sum_{i=1}^{n} \mathsf{X}_n \\ M_n &= \frac{S_n}{n} \end{aligned}$$

- Important: Different realisations of n samples will result in different M_n. Thus M_n is a random variable.
- Natural question: What happens to M_n as n becomes large
- Intuitive answer: It becomes closer to μ.
- What does this mean?

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Law of Large Numbers

- As a first approximation this means that the variance of M_n decreases with increasing n.
- Recall that the variance of $\sum_{i=1}^{n} X_n$ will be $n\sigma^2$ and hence the variance of M_n would be $\frac{1}{n^2}\ n\sigma^2 = \frac{\sigma^2}{n}$.
- This implies M_n is more likely to be close to μ.
- This is the Law of Large Numbers

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Central Limit Theorem

- Let us now consider a slightly different "scaling" and shifting
- Recall that $\mathsf{E}(S_n - n\mu) = 0$ and

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Central Limit Theorem

- Let us now consider a slightly different "scaling" and shifting
- Recall that $\mathsf{E}(S_n - n\mu) = 0$ and

$$Z_n := \frac{S_n - n\mu}{\sqrt{n}\sigma}$$

- As n becomes large Z_n 'approaches' a unit normal random variable.
- This means that for 'large-ish' n Z_n can be approximated by a unit normal.

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Limit Theorems

- LLN and CLT are called limit theorems.
- They describe how random variables behave ‘in the aggregate’
- LLN tells us that the sample average gets us close to the true average
- CLT tells us how close we get to the true average
- It motivates the methods used for polling, for developing insurance schemes, for doing simulations,
- CLT justifes the widespread use of, and interest in, the Gaussian distribution.

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Were the polls wrong

SEARCH

The New York Times

TheUpshot

Alaska Ariz. Colo. Fla. Ga. Ind. Iowa Miss. Mo. N.C. N.H. Nev. Ohio Pa. Wis.

Who Will Be President?

By JOSH KATZ

The estimates on this page are based on pre-election polls. For an estimate including results, The T

President **Senate** **House**

Hillary Clinton has an **85% chance** to win.



Last updated Tuesday, November 8 at 10:20 PM ET

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Were the polls wrong

Race	Poll model	Hillary Clinton Democratic	Donald Trump Republican	Gary Johnson Libertarian	Jill Stein Green	Leading by (points)
Two-way	270toWin[4]	**47.2%**	43.6%	—		**3.6**
	BBC[5]	**48.0%**	44.0%			**4.0**
	HuffPost Pollster[6]	**47.3%**	42.0%			**5.3**
	New York Times[7]	**45.9%**	42.8%			**3.1**
	Real Clear Politics[8]	**46.8%**	43.6%			**3.2**
	TPM Polltracker[9]	**48.8%**	43.9%			**4.9**
Three-way	FiveThirtyEight[10]	**45.7%**	41.8%	4.8%	—	**3.9**
	HuffPost Pollster[6]	**45.7%**	40.8%	5.0%		**4.9**
	New York Times[7]	**45.4%**	42.3%	5.0%		**3.1**
	TPM Polltracker[9]	**46.0%**	44.1%	4.9%		**1.9**
Four-way	270toWin[4]	**45.6%**	42.5%	4.8%	2.1%	**3.1**
	Real Clear Politics[11]	**45.5%**	42.2%	4.7%	1.9%	**3.3**
	CNN Poll of Polls[12]	**46.0%**	42.0%	5.0%	2.0%	**4.0**
	TPM Polltracker[9]	**46.6%**	43.8%	4.6%	2.7%	**2.8**
Election results (popular vote)		**48.2%**	46.1%	3.3%	1.1%	**2.1**

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Estimators: Point and Interval Estimate

- Consider a population. Interest is to estimate a parameter associated with the population.
- First step is to obtain n **random** samples from the population and measure the parameter of interest.
- Let X_i be the random variable denoting the possible value from the i-th sample.
- Let θ be the parameter of interest.
- Let $x = \{x_1, x_2, \ldots, x_n\}$ be values from the sample measurements
- x_i is *realisation* of random variable X_i.
- Any function of the samples x is called a *statistic*.

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Estimators: Point and Interval Estimate

- $\hat{\theta} = g(x_1, x_2, \ldots, x_n)$ is a *point estimate* of θ.
- $\left(\mathsf{E}\left(\hat{\theta}\right) - \theta\right)$ is called the **bias** of the estimate.
- $\mathsf{E}\left(\left(\hat{\theta} - \theta\right)^2\right)$ is called the mean squared error of the estimate.
- An alterntaive to a point estimate is the *interval estimate* described using $\theta_l, \theta_u, \gamma$

$$\mathsf{Prob}(\theta_l \leq \theta \leq \theta_u) \geq \gamma$$

- γ is called the *confidence level* and (θ_l, θ_u) is the the *interval estimate.*

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Recap: Sums of Independent Random Variables

- Consider the sum of n independent and identically distributed (iid) random variables—X_i, $i = 1, \ldots n$.
- Let $\mathsf{E}(\mathsf{X}_1) = \mu$ and $\mathsf{VAR}(\mathsf{X}_1) = \sigma^2$.
- Let the sample sum be $\mathsf{S}_n := \sum_{i=1}^{n} \mathsf{X}_i$.
- We know that $\mathsf{E}(\mathsf{S}_n) = n\mu$ and $\mathsf{VAR}(\mathsf{S}_n) = n\sigma^2$.
- From the Law of Large Numbers, $\frac{\mathsf{S}_n}{n} \longrightarrow \mu$,
  - Intuitively, if we take a large number of samples, the sample mean will 'approach the true mean.

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Recap: Sums of Independent Random Variables

- From Central Limit Theorem, with Z being the unit Normal,

$$\text{Prob}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} < y\right) \rightarrow \text{Prob}(Z < y)$$

  - As n becomes large, the distribution of the sum approaches that of the Gaussian distribution with same mean and same variance as the sum.
  - Thus for large n, the Gaussian is a good approximation for the distribution of the sum of n iid random variables.

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The Unit Normal

- Z is the zero-mean Gaussian random variable with unit variance.
- If X is a Gaussian random variable with mean μ and standard deviation σ, then $\frac{\mathsf{X}-\mu}{\sigma}$ is unit normal.
- The above operation essentially *shifts and scales* the original Gaussian, it normalises it.
- Some 'unit normal' probabilities
  - $\mathsf{Prob}(|\mathsf{Z}| < 1) \approx 0.68$
  - $\mathsf{Prob}(|\mathsf{Z}| < 2) \approx 0.95$
  - $\mathsf{Prob}(|\mathsf{Z}| < 3) \approx 0.997$
  - $\mathsf{Prob}(|\mathsf{Z}| > 1) \approx 1 - 0.68 = 0.32$

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The Unit Normal

- Prob$(0 \leq \mathsf{Z} < 1) \approx ?$
- Prob$(\mathsf{Z} \notin (0, 1)) \approx ?$
- Prob$(\mathsf{Z} \notin (0, 2)) \approx ?$
- Prob$(\mathsf{Z} > z_\alpha) = \alpha$

Percentile	90	95	97.5	99	99.5	99.9	99.95
α	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
z_α	1.28	1.645	1.96	2.33	2.58	3.08	3.27

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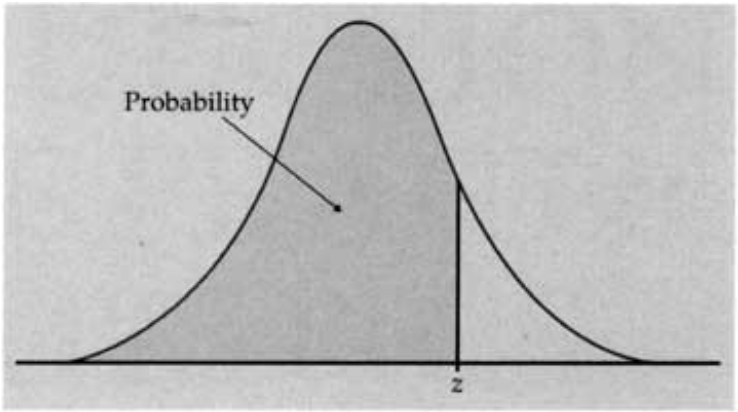


Table entry for z is the area under the standard normal curve to the left of z.

TABLE A Standard normal probabilities (*continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

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Sample Mean

- Assume a population with mean μ and variance σ^2.
  - μ is unknown and is to be estimated, σ is known.
  - Neither is known and are to be estimated.
- 'Obvious method': Obtain n independent samples whose values will be $X_1, \ldots X_n$ and calculate the sample mean as an estimator for the mean.

$$\text{Sample Mean} := \hat{\mu}_n := \frac{\sum_{i=1}^{n} X_i}{n}$$

- Clearly, $\hat{\mu}_n$ is a random variable. Characterising this random variable characterises the quality of the estimate.

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Sample Mean

- The expectation and variance of the sample mean are

$$\mathsf{E}(\hat{\mu}_n) = \mathsf{E}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right)$$

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Sample Mean

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$$\mathsf{E}(\hat{\mu}_n) = \mathsf{E}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i)}{n}$$

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Sample Mean

- The expectation and variance of the sample mean are

$$\mathsf{E}(\hat{\mu}_n) = \mathsf{E}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i)}{n} = \mu$$

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Sample Mean

- The expectation and variance of the sample mean are

$$\begin{aligned} \mathsf{E}(\hat{\mu}_n) &= \mathsf{E}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^{n} \mathsf{E}(\mathsf{X}_i)}{n} = \mu \\ \mathsf{VAR}(\hat{\mu}_n) &= \mathsf{VAR}\left(\frac{\sum_{i=1}^{n} \mathsf{X}_i}{n}\right) \end{aligned}$$

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Sample Mean

- The expectation and variance of the sample mean are

$$\begin{aligned} \mathsf{E}(\hat{\mu}_n) &= \mathsf{E}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{E}(\mathsf{X}_i)}{n} = \mu \\ \mathsf{VAR}(\hat{\mu}_n) &= \mathsf{VAR}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{VAR}(\mathsf{X}_i)}{n} \end{aligned}$$

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Sample Mean

- The expectation and variance of the sample mean are

$$\begin{aligned} \mathsf{E}(\hat{\mu}_n) &= \mathsf{E}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{E}(\mathsf{X}_i)}{n} = \mu \\ \mathsf{VAR}(\hat{\mu}_n) &= \mathsf{VAR}\left(\frac{\sum_{i=1}^n \mathsf{X}_i}{n}\right) = \frac{\sum_{i=1}^n \mathsf{VAR}(\mathsf{X}_i)}{n} = \frac{\sigma^2}{n} \end{aligned}$$

- What is the distribution of $\hat{\mu}_n$?
- How about 'an approximate distribution' of $\hat{\mu}_n$?

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^{n} (X_i - \mu)^2$$

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} V &= \sum_{i=1}^{n} (X_i - \mu)^2 \\ E(V) &= E\left(\sum_{i=1}^{n} (X_i - \mu)^2 \right) \end{aligned}$$

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \mu)^2 \\
E(V) &= E\left(\sum_{i=1}^{n} (X_i - \mu)^2 \right) \\
&= \sum_{i=1}^{n} E\left((X_i - \mu)^2 \right)
\end{aligned}
$$

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} \mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2 \\ \mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2 \right) \\ &= \sum_{i=1}^{n} \mathsf{E}\left((\mathsf{X}_i - \mu)^2 \right) \\ &= n\sigma^2 \end{aligned}$$

- Thus if μ is known then V/n is an unbiased estimator of $\mathsf{VAR}(\mathsf{X}_1)$.

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^{n} (X_i - \hat{\mu})^2$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\ \sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \end{aligned}$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \\
E(V) &= E\left(\sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \right)
\end{aligned}
$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
\mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 &= \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \\
\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}\left(\mathsf{X}_i^2\right) - n\mathsf{E}\left((\hat{\mu})^2\right)
\end{aligned}
$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

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\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \\
E(V) &= E\left(\sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} E\left(X_i^2\right) - nE\left((\hat{\mu})^2\right) \\
&= nVAR(X_1) + n(E(X_1))^2 - nVAR(\hat{\mu}) - n(E(\hat{\mu}))
\end{aligned}
$$

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- Consider the sum of the square of the deviations from the mean, i.e.,

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\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}\left(\mathsf{X}_i^2\right) - n\mathsf{E}\left((\hat{\mu})^2\right) \\
&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu})) \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2
\end{aligned}
$$

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Sample Variance: Unknown mean

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\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}\left(\mathsf{X}_i^2\right) - n\mathsf{E}\left((\hat{\mu})^2\right) \\
&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu})) \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \; =
\end{aligned}
$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

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\mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 \\
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\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}\left(\mathsf{X}_i^2\right) - n\mathsf{E}\left((\hat{\mu})^2\right) \\
&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu})) \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \;=\; (n-1)\sigma^2.
\end{aligned}
$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
\mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 &= \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \\
\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}\left(\mathsf{X}_i^2\right) - n\mathsf{E}\left((\hat{\mu})^2\right) \\
&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu})) \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \;=\; (n-1)\sigma^2.
\end{aligned}
$$

- Thus, when μ is not known, $\frac{V}{n-1}$ is the unbiased estimate of $\mathsf{VAR}(\mathsf{X}_1)$.

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Lecture 3

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Recap

- Random independent samples from a population
- Each sample can is an independent random variable
- Sample mean is a statistic.
- Expectation of sample mean is the same as the true mean.
- Variance of the sample mean is σ^2/n
- Since samples are independent, we can invoke LLN and CLT on the Sample Sum.

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^{n} (X_i - \mu)^2$$

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} V &= \sum_{i=1}^{n} (X_i - \mu)^2 \\ E(V) &= E\left(\sum_{i=1}^{n} (X_i - \mu)^2 \right) \end{aligned}$$

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
V &= \sum_{i=1}^{n} (X_i - \mu)^2 \\
E(V) &= E\left(\sum_{i=1}^{n} (X_i - \mu)^2 \right) \\
&= \sum_{i=1}^{n} E\left((X_i - \mu)^2 \right)
\end{aligned}
$$

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Sample Variance: Known mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} \mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2 \\ \mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} (\mathsf{X}_i - \mu)^2 \right) \\ &= \sum_{i=1}^{n} \mathsf{E}\left((\mathsf{X}_i - \mu)^2 \right) \\ &= n\sigma^2 \end{aligned}$$

- Thus if μ is known then V/n is an unbiased estimator of $\mathsf{VAR}(\mathsf{X}_1)$.

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$V = \sum_{i=1}^{n} (X_i - \hat{\mu})^2$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$\begin{aligned} V &= \sum_{i=1}^{n} (X_i - \hat{\mu})^2 \\ \sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \end{aligned}$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

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Sample Variance: Unknown mean

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\sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \\
E(V) &= E\left(\sum_{i=1}^{n} X_i^2 - n(\hat{\mu})^2 \right)
\end{aligned}
$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

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\begin{aligned}
\mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 &= \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \\
\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}\left(\mathsf{X}_i^2\right) - n\mathsf{E}\left((\hat{\mu})^2\right)
\end{aligned}
$$

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Sample Variance: Unknown mean

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\sum_{i=1}^{n} (X_i - \hat{\mu})^2 &= \sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \\
E(V) &= E\left(\sum_{i=1}^{n} X_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} E\left(X_i^2\right) - nE\left((\hat{\mu})^2\right) \\
&= nVAR(X_1) + n(E(X_1))^2 - nVAR(\hat{\mu}) - n(E(\hat{\mu}))
\end{aligned}
$$

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Sample Variance: Unknown mean

- Consider the sum of the square of the deviations from the mean, i.e.,

$$
\begin{aligned}
\mathsf{V} &= \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 \\
\sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2 &= \sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \\
\mathsf{E}(\mathsf{V}) &= \mathsf{E}\left(\sum_{i=1}^{n} \mathsf{X}_i^2 - n\,(\hat{\mu})^2 \right) \\
&= \sum_{i=1}^{n} \mathsf{E}\left(\mathsf{X}_i^2\right) - n\mathsf{E}\left((\hat{\mu})^2\right) \\
&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu})) \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2
\end{aligned}
$$

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Sample Variance: Unknown mean

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&= \sum_{i=1}^{n} \mathsf{E}\left(\mathsf{X}_i^2\right) - n\mathsf{E}\left((\hat{\mu})^2\right) \\
&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu})) \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \; =
\end{aligned}
$$

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Sample Variance: Unknown mean

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&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu})) \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \;=\; (n-1)\sigma^2.
\end{aligned}
$$

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&= n\mathsf{VAR}(\mathsf{X}_1) + n(\mathsf{E}(\mathsf{X}_1))^2 - n\mathsf{VAR}(\hat{\mu}) - n(\mathsf{E}(\hat{\mu})) \\
&= n\sigma^2 + n\mu^2 - n\left(\frac{\sigma^2}{n}\right) - n\mu^2 \;=\; (n-1)\sigma^2.
\end{aligned}
$$

- Thus, when μ is not known, $\frac{V}{n-1}$ is the unbiased estimate of $\mathsf{VAR}(\mathsf{X}_1)$.

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CLT and the Sample Mean

- Let μ and σ^2 be the true mean and true variance of a population feature.
- Applying CLT to the sample sum of the n independent sample, S_n we have

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} \rightarrow \mathsf{Z}$$

  where Z is the unit normal.
- Let $\mathsf{Prob}(\mathsf{Z} > z_\alpha) = \alpha$

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CLT and the Sample Mean

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$$\frac{S_n - n\mu}{\sqrt{n}\sigma} \rightarrow \mathsf{Z}$$

  where Z is the unit normal.
- Let $\mathsf{Prob}(\mathsf{Z} > z_\alpha) = \alpha$

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} = n\frac{S_n/n - \mu}{\sqrt{n}\sigma}$$

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CLT and the Sample Mean

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- Let $\mathsf{Prob}(\mathsf{Z} > z_\alpha) = \alpha$

$$\frac{S_n - n\mu}{\sqrt{n}\sigma} = n\frac{S_n/n - \mu}{\sqrt{n}\sigma} = \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})}$$

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CLT and the Sample Mean

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CLT and the Sample Mean

- Let us now characterize deviation of sample mean from true mean.

$$\text{Prob}\left(-z_\alpha \leq \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})} \leq z_\alpha\right) = 1 - 2\alpha$$

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CLT and the Sample Mean

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$$\text{Prob}\left(-z_\alpha \leq \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})} \leq z_\alpha\right) = 1 - 2\alpha$$

$$\text{Prob}\left(-\frac{\sigma}{\sqrt{n}} z_\alpha \leq (\hat{\mu}_n - \mu) \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) = 1 - 2\alpha$$

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CLT and the Sample Mean

- Let us now characterize deviation of sample mean from true mean.

$$\begin{aligned}
\text{Prob}\left(-z_\alpha \leq \frac{\hat{\mu}_n - \mu}{(\sigma/\sqrt{n})} \leq z_\alpha\right) &= 1-2\alpha \\
\text{Prob}\left(-\frac{\sigma}{\sqrt{n}} z_\alpha \leq (\hat{\mu}_n - \mu) \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha \\
\text{Prob}\left(|\hat{\mu}_n - \mu| \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha
\end{aligned}$$

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CLT and the Sample Mean

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\text{Prob}\left(-\frac{\sigma}{\sqrt{n}} z_\alpha \leq (\hat{\mu}_n - \mu) \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha \\
\text{Prob}\left(|\hat{\mu}_n - \mu| \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha \\
\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}} z_\alpha \leq \mu \leq \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha
\end{aligned}$$

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CLT and the Sample Mean

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\text{Prob}\left(-\frac{\sigma}{\sqrt{n}} z_\alpha \leq (\hat{\mu}_n - \mu) \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha \\
\text{Prob}\left(|\hat{\mu}_n - \mu| \leq z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha \\
\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}} z_\alpha \leq \mu \leq \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= 1-2\alpha
\end{aligned}$$

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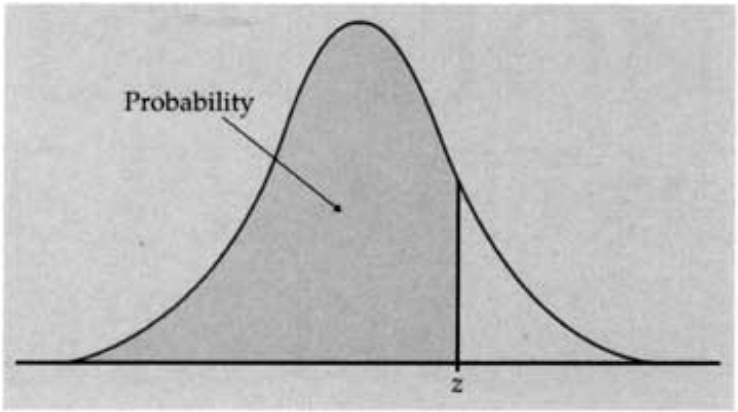


Table entry for z is the area under the standard normal curve to the left of z.

TABLE A Standard normal probabilities (*continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

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CLT and the Sample Mean

$$\text{Prob}\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}}\, z_\alpha \;\leq\; \mu \;\leq\; \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right) \quad = \quad 1 - 2\alpha$$

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- $\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}}\, z_\alpha, \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right)$ is called the $(1-\alpha)$ confidence interval for μ.
- Usually, variance is not known!

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- $\left(\hat{\mu}_n - \frac{\sigma}{\sqrt{n}}\, z_\alpha, \hat{\mu}_m + z_\alpha \frac{\sigma}{\sqrt{n}}\right)$ is called the $(1-\alpha)$ confidence interval for μ.
- Usually, variance is not known! So ...
- Compute

$$\hat{V} = \frac{1}{n-1} \sum_{i=1}^{n} (\mathsf{X}_i - \hat{\mu})^2$$

and use $\hat{V}$ as the variance, i.e., use $\sigma = \sqrt{\hat{V}}$ in the preceding.

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The $n\sigma$ rules

- 3σ and 6σ rule is something that is commonly used in quality control departments.
- Most manufacturing processes have tolerance; the allowable deviation from the nominal value.
- Production outside of this tolerance band is discarded;
- A 3σ production process means that deviation from the nominal value are such that 3σ is within the tolerance ban.
- Lower tolerance requires correspondingly lower σ and a tighter control over the process.

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Some experiments

- Write a program to do the following:
  1. Generate n IID random variables of a specified distribution F whose mean and standard deviation are known.
  2. Compute the sample mean and the limits of the 95% confidence interval.
  3. Check if the true mean is in the 95% confidence interval;
  4. In an outer loop, repeat this experiment K times.
  5. Plot the histogram of the sample mean from the K experiments.
  6. Make a scatter plot of the left and the right limits of the 95% confidence interval.

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Some experiments

- Let F correspond to Gaussian with $\mu = 0$ and $\sigma = 2$. Let $n = 20$ and $K = 1000$. Repeat for different $\sigma = 3, 5$ and $n = 50, 100, 200, 500, 1000$.
- Now do it for F uniformly distributed in $(0, 1)$, $(-1, 1)$, and $(-2, 2)$.
- And for F being a Bernouilli distribution
- Now pretend σ is not known and use $\hat{V}$ as above and see if there is a difference.

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Parameter Estimation

- Many times, we have a model for the CDF of pdf of population feature of interest.
- The model CDF will have parameters.
- We will now discuss how to determine the **best estimate** of the paramter(s).
- As before we will obtain n independent sample realizations of the parameter of interest.

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What is being estimated?



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What is being estimated?



And what is the method?

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Likelihood Function

- $x_1, \ldots x_K$ is the set of measurements.
- Assume that the parameter set is θ.
- Question: What is the *likelihood* of observing of $x_1, \ldots x_K$ if the parameter value was θ?
  - Assume the population is Bernoulli with parameter p.
  - Assume the population is Poisson with parameter λ.
  - Assume the population is uniform in the interval $(0, a)$.
  - Assume the population is Gaussian with unit variance and mean μ.
  - Assume the population is exponential with parameter μ.

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Maximum Likelihood Estimate

- Find the θ that maximises the likelihood of the observations.

$$\hat{\theta} = \arg\max_{\theta} L(x_1, \ldots, x_K|\theta)$$

- Can also maximise the Log Likelihood function.
- Likelihood and log likelihood functions and MLEs
- Bernouilli

$$\begin{aligned} L(\underline{x}; p) &= \prod_{i=1}^{K} p^{x_i}(1-p)^{1-x_i} = p^n(1-p)^{K-n} \\ \hat{p} &= \frac{\sum_{i=1}^{K} x_i}{K} \end{aligned}$$

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